



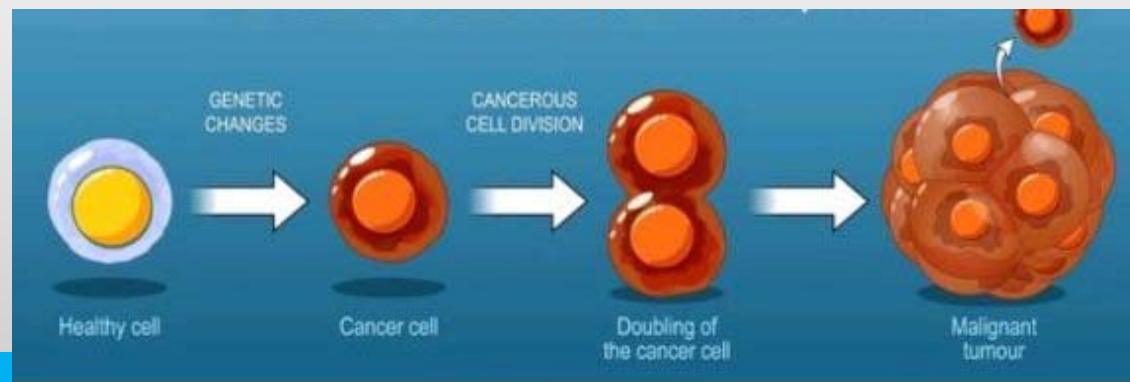
Neoplasia

Lec. 11

Dr.Zainab Hashim

Neoplasia

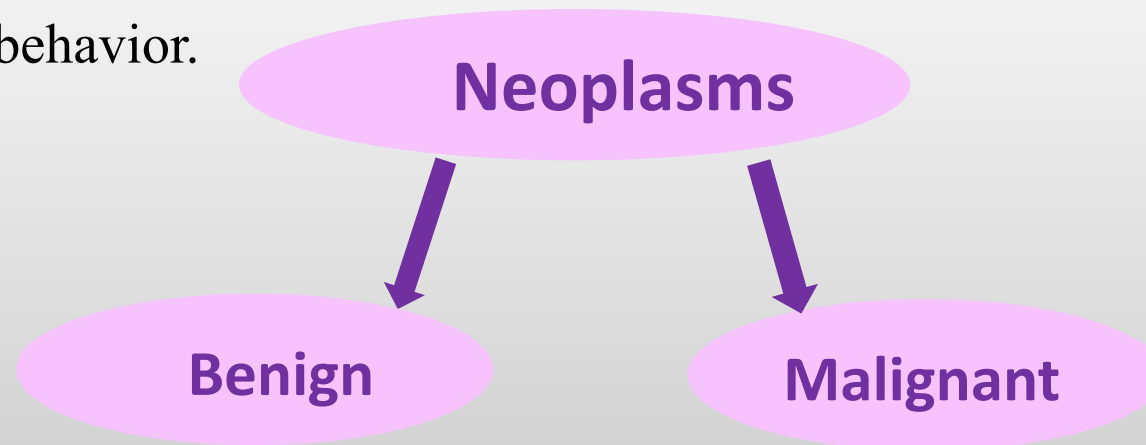
An abnormal tissue proliferation, which exceeds that of adjacent normal tissue. This proliferation continues even after removing the causative agent. Neoplastic cells are said to be transformed because they continue to replicate, apparently oblivious to the regulatory influences that control normal cells. Neoplasms therefore enjoy a degree of autonomy and tend to increase in size regardless of their local environment.



Neoplasia literally means a new growth.

Oncology the study of tumors (from oncos, “tumor,” and logos, “study of”)

- Neoplasm arises from mutations in genes that regulate **cell growth**, **death** or **DNA repair**. Collectively the genetic alterations in tumor cells provide them growth and survival advantages over normal cells.
- The neoplasms divide into benign and malignant categories based on a tumor’s potential clinical behavior.



Benign tumors

- Benign tumors have microscopic and gross characteristics considered to be relatively innocent, implying that it will remain localized and is amenable to local surgical removal. Affected patients generally survive.
- In general, benign tumors are named by attaching the **suffix -oma** to the cell of origin . For example, a benign tumor arising in fibrous tissue is a fibroma; a benign cartilaginous tumor is a chondroma .

Malignant tumors

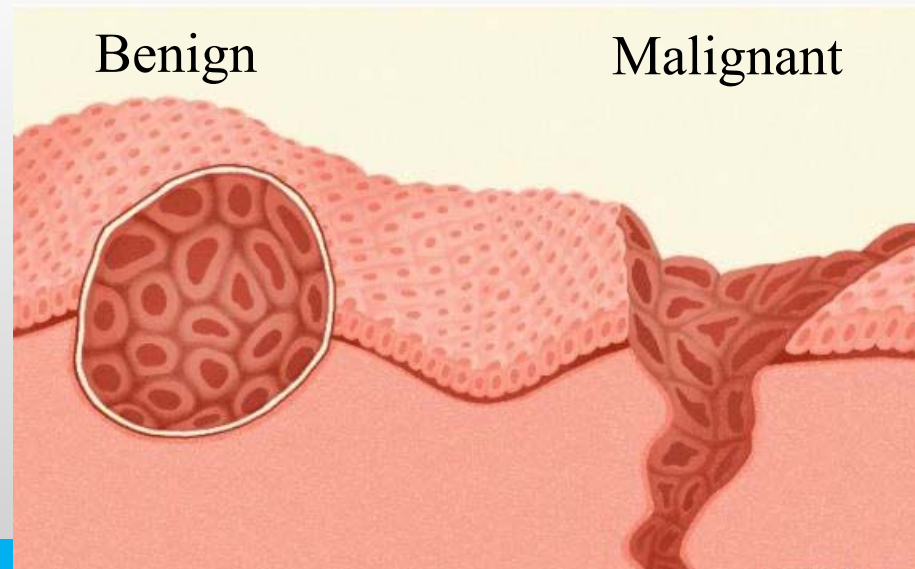
- Malignant, as applied to a neoplasm, implies that the lesion can invade and destroy adjacent structures and spread to distant sites (metastasize) to cause death. Malignant tumors are collectively referred to as cancers, derived from the Latin word for “crab”—that is, they adhere to any part that they seize in an obstinate manner, similar to a crab’s behavior.



- Malignant tumors that arise from or differentiating towards mesenchymal cells are named by attaching the **suffix - sarcomas**. Examples include fibrosarcoma (fibroblasts), liposarcoma (lipocytes).
- While the malignant neoplasms of epithelial cells are called **carcinomas** . Examples include : squamous cell carcinoma and renal cell carcinoma .

Characteristics of benign and malignant neoplasms

- There are three fundamental features by which most benign and malignant tumors can be distinguished:
 1. Differentiation and anaplasia
 2. Local invasion
 3. Metastasis



Differentiation and anaplasia

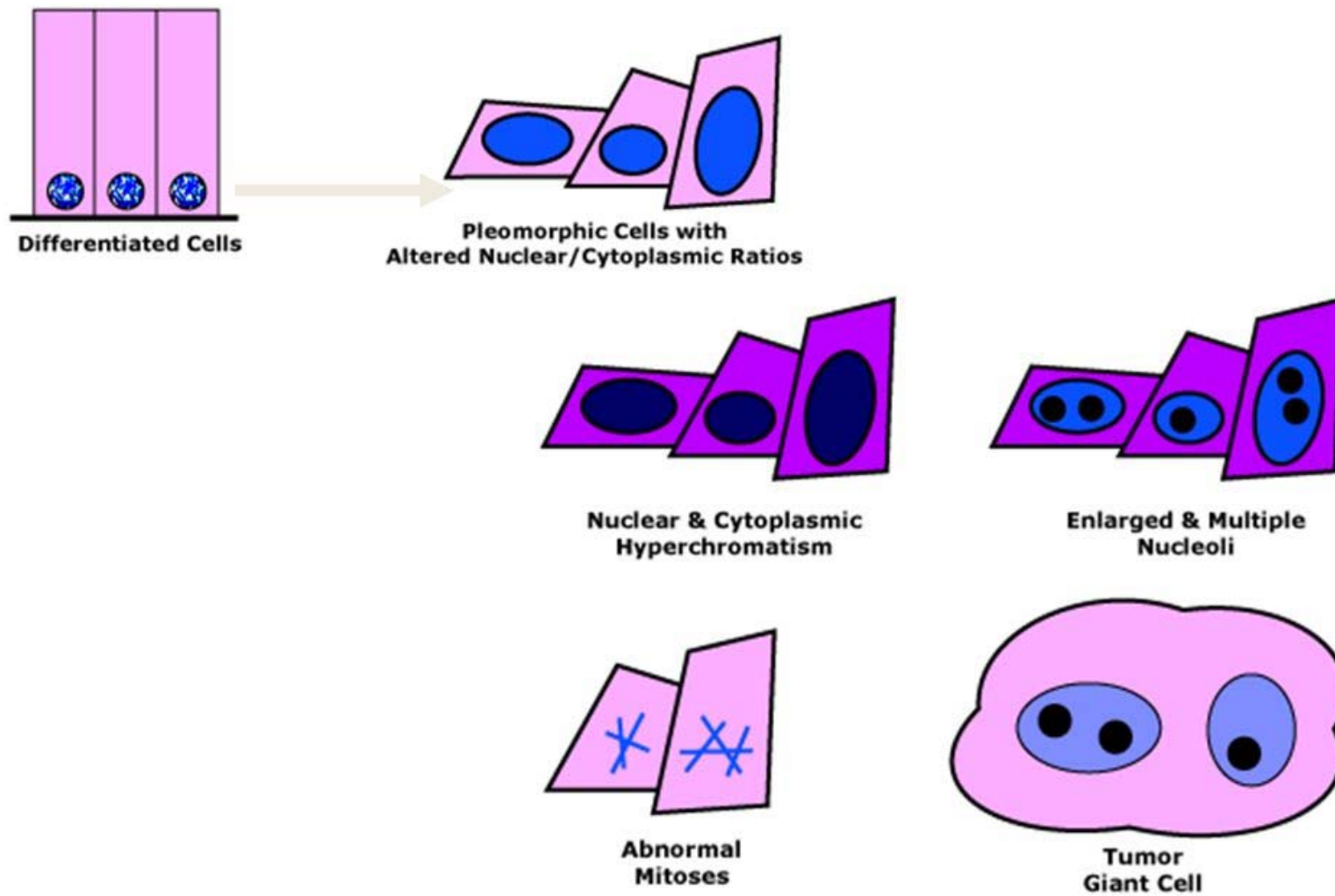
- Differentiation refers to the extent to which neoplasms resemble their parenchymal cells of origin, both morphologically and functionally; lack of differentiation is called **anaplasia**.
- In general, benign neoplasms are composed of well-differentiated cells that closely resemble their normal counterparts while malignant neoplasms exhibit a wide range of parenchymal cell differentiation.

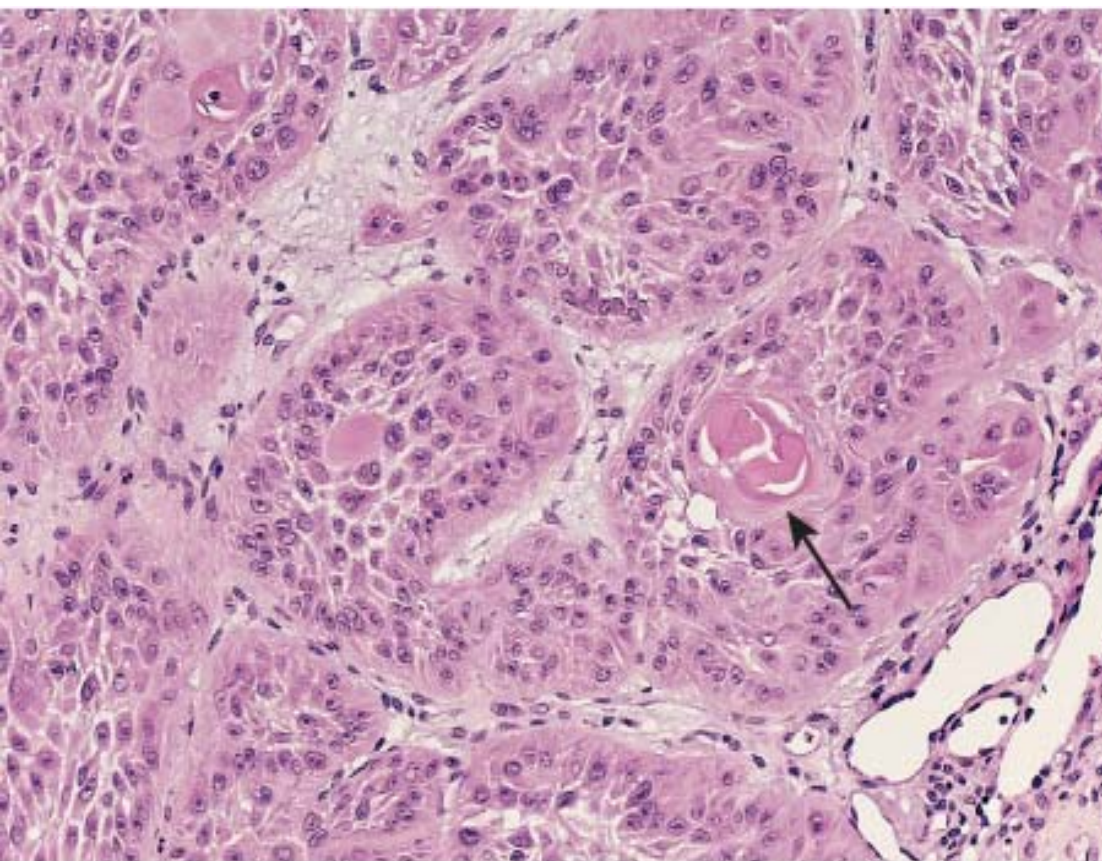
- Malignant neoplasms are generally divided into four categories depending on their degree of differentiation, these are :
- Well-differentiated neoplasms
- Poorly differentiated neoplasms
- Moderately differentiated neoplasms
- Undifferentiated (anaplastic) neoplasms

- Tumors composed of undifferentiated cells are said to be **anaplastic**, a feature that is a reliable indicator of malignancy. Anaplastic cells often display the following morphologic features:

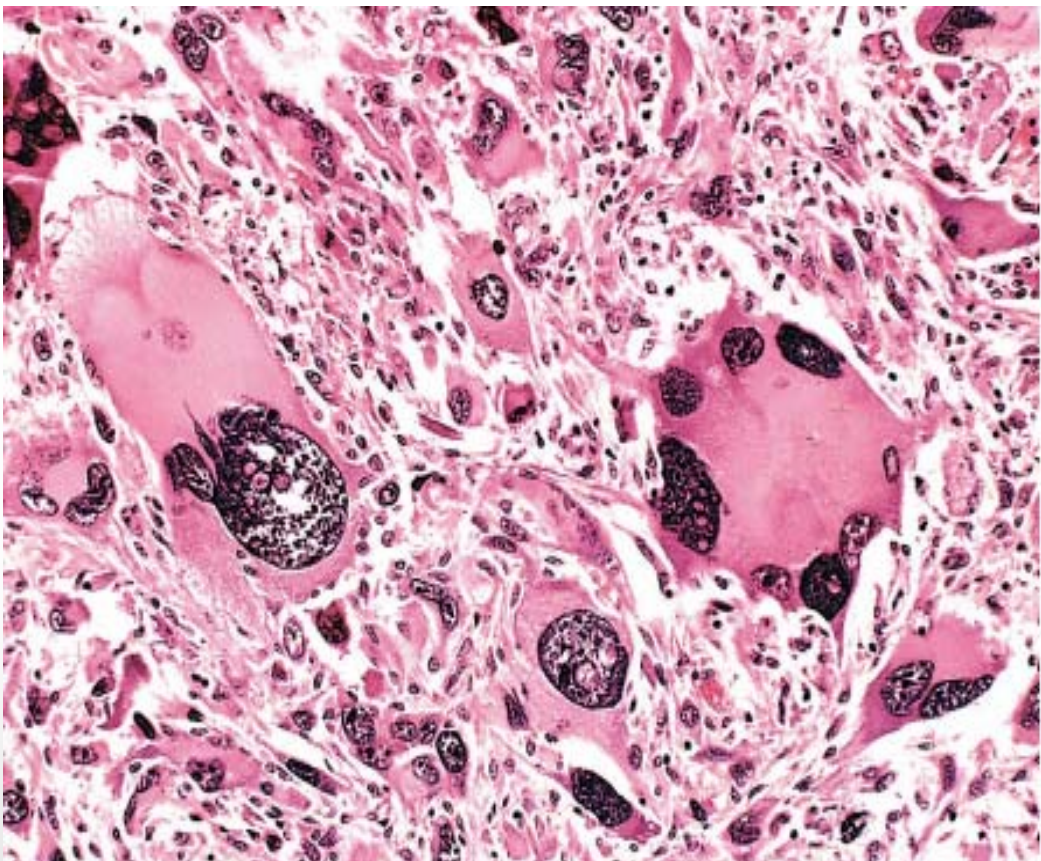
1. **Pleomorphism** (variation in size and shape) .
2. **Nuclear abnormalities** consisting of extreme hyperchromatism (dark-staining), variation in nuclear size and shape, or unusually prominent single or multiple nucleoli.
3. **Tumor giant cells** may be formed. These are considerably larger than neighboring cells and may possess either one enormous nucleus or several nuclei.
4. **Atypical mitoses**, which may be numerous.
5. **Loss of polarity**, such that anaplastic cells lack recognizable patterns of orientation to one another.

Histologic Features of Malignant Cells





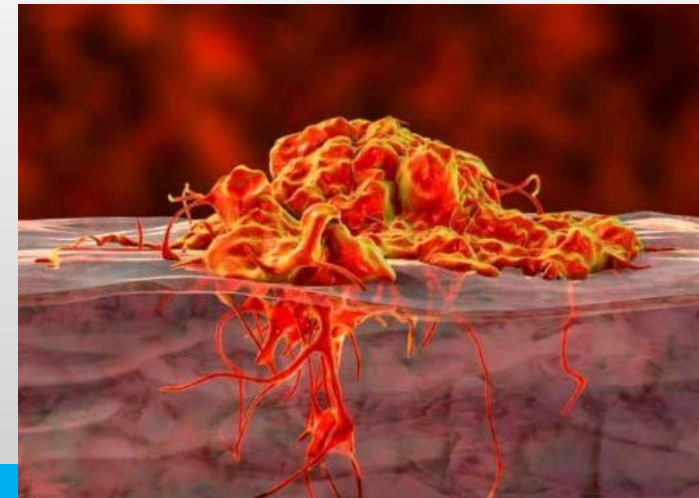
Well-differentiated squamous cell carcinoma of the skin. The tumor cells are strikingly similar to normal squamous epithelial cells, with intercellular bridges and nests of keratin (arrow).



Pleomorphic malignant tumor (rhabdomyosarcoma). Note the marked variation in cell and nuclear sizes, the hyperchromatic nuclei, and the presence of tumor giant cells.

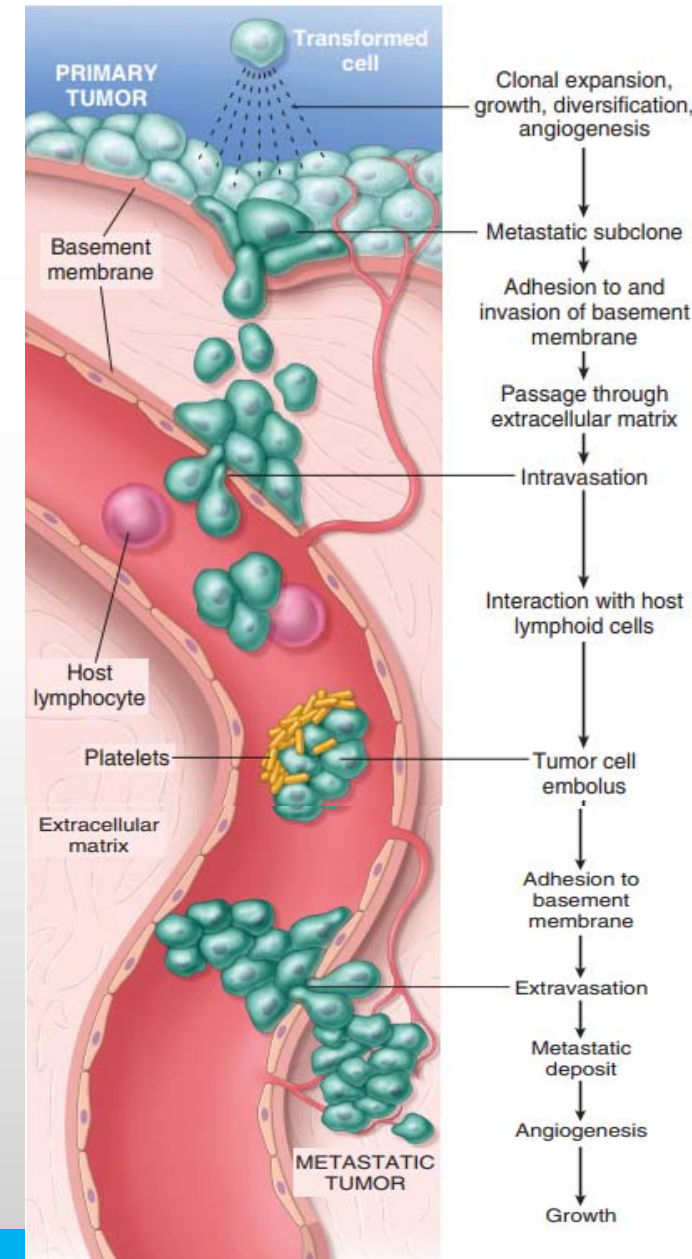
Local Invasion

- The growth of cancers is accompanied by progressive infiltration, invasion, and destruction of surrounding tissues, whereas most benign tumors grow as cohesive expansile masses that remain localized to their sites of origin. Because benign tumors grow and expand slowly, they usually develop a rim of compressed fibrous tissue. This capsule consists largely of extracellular matrix.



Metastasis

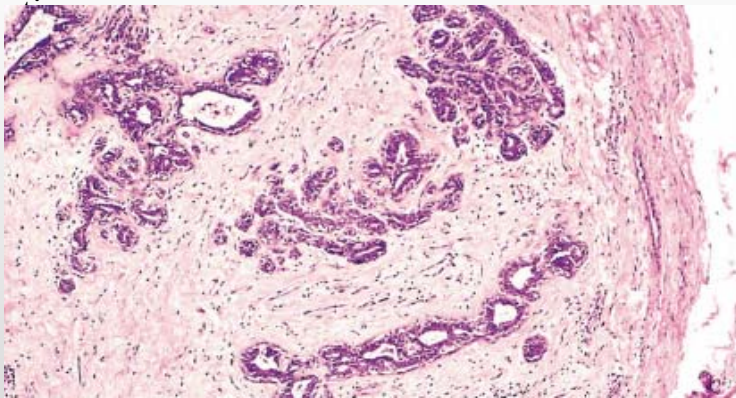
- Metastasis is defined by the spread of a tumor to sites that are physically discontinuous with the primary tumor and marks a tumor as malignant, while benign neoplasms do not metastasize. The invasiveness of cancers permits them to penetrate into blood vessels, lymphatics, and body cavities, providing opportunities for spread.



Benign tumor

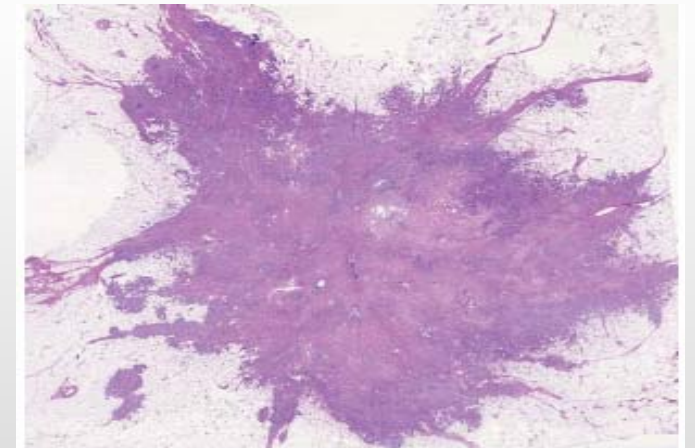
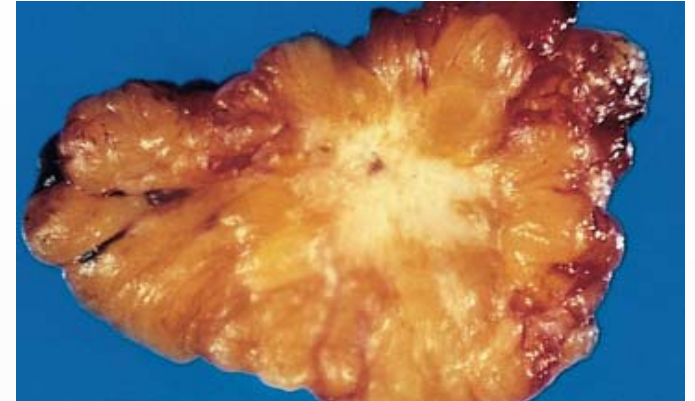


Fibroadenoma of the breast. The tan-colored, encapsulated small tumor is sharply demarcated from the whiter breast



Microscopic view of fibroadenoma of the breast . The fibrous capsule (right) sharply delimits the tumor from the surrounding tissue.

Malignant tumor



invasive ductal carcinoma of the breast. The lesion is retracted, infiltrating the surrounding breast substance

Benign tumors

- Resemble the tissue of origin and are well differentiated
- Tend to be slow growing
- Well circumscribed and have a capsule
- Remain localized to the site of origin

Malignant Tumors

- Poorly or completely undifferentiated (anaplastic).
- Grow faster.
- Poorly circumscribed and invade the surrounding normal tissues.
- Locally invasive and metastasize to distant sites.

Etiology of cancer

- Genetic damage lies at the heart of carcinogenesis. Three classes of gene damaging or carcinogenic agents can be identified:
 1. Chemicals
 2. Radiant energy
 3. Microbial agents

Thank You

